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Oscillator circuit

Abstract

An oscillator circuit includes: an inverter; a crystal oscillator coupled in parallel to the inverter; a first load capacitor coupled to an input portion of the crystal oscillator; a second load capacitor coupled to an output portion of the crystal oscillator; and a band stop filter coupled in parallel to the inverter and having a rejection band including a resonant frequency of the crystal oscillator. Also, the band stop filter is a twin-T notch filter.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2023-107981, filed Jun. 30, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an oscillator circuit.

2. Related Art

(3) Crystal oscillator circuits described in JP-A-54-061447 and JP-A-08-023230 each include a corresponding CMOS inverter coupled between a power source and ground. Also, a series circuit including an output resistor and a crystal oscillator, and a DC feedback resistor are coupled in parallel between the input end of the inverter and the output end thereof. Also, one end of the crystal oscillator is coupled to one end of a first load capacitor, and the other end of the crystal oscillator is coupled to one end of a second load capacitor. In JP-A-54-061447, the other ends of the first and the second load capacitors are coupled to the power source. In JP-A-08-023230, the other ends of the first and the second load capacitors are coupled to ground.

(4) However, with the crystal oscillator circuits described in JP-A-54-061447 and JP-A-08-023230, noise is likely to increase due to their configurations.

SUMMARY

(5) According to an aspect of the present disclosure, there is provided an oscillator circuit including: an inverter; a crystal oscillator coupled in parallel to the inverter; a first load capacitor coupled to an input portion of the crystal oscillator; a second load capacitor coupled to an output portion of the crystal oscillator; and a band stop filter coupled in parallel to the inverter and having a rejection band including a resonant frequency of the crystal oscillator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a circuit diagram of an oscillator circuit according to a first embodiment.
(2) FIG. **2** is an exploded perspective view of an accelerometer.
(3) FIG. **3** is a graph illustrating the characteristics of a twin-T notch filter.
(4) FIG. **4** is a circuit diagram of an oscillator circuit according to a second embodiment.
(5) FIG. **5** is a graph illustrating the characteristics of a twin-T notch filter.
(6) FIG. **6** is a diagram illustrating a waveform of an oscillation signal.
(7) FIG. **7** is a diagram illustrating a waveform of an oscillation signal.
DESCRIPTION OF EMBODIMENTS
(8) In the following, a detailed description will be given of an oscillator circuit according to the present disclosure in accordance with embodiments described with reference to the attached drawings.
First Embodiment
(9) FIG. **1** is a circuit diagram of the oscillator circuit according to a first embodiment. FIG. **2** is an exploded perspective view of an accelerometer. FIG. **3** is a graph illustrating the characteristics of a twin-T notch filter.
(10) An oscillator circuit **100** illustrated in FIG. **1** is a Colpitts type oscillator circuit that oscillates at a predetermined frequency and includes a crystal oscillator **110**, an inverter **120**, a feedback resistor **130**, a limiting resistor **140**, a first load capacitor **150**, and a second load capacitor **160**.
(11) Among these elements, the crystal oscillator **110**, the inverter **120**, and the feedback resistor **130** are coupled in parallel with each other. The limiting resistor **140** is coupled between the output end of the inverter **120** and the input end of the crystal oscillator **110**. The first load capacitor **150** is coupled between the input end of the crystal oscillator **110** and ground. The second load capacitor **160** is coupled between the output end of the crystal oscillator **110** and ground. In this regard, the first load capacitor **150** and the second load capacitor **160** may be coupled to a power source VDD instead of ground.
(12) The crystal oscillator **110** is mounted on an accelerometer **200**. Accordingly, before specifically describing the oscillator circuit **100**, a brief description will be given of the accelerometer **200** with reference to FIG. **2**. In this regard, for convenience of explanation, an X-axis, a Y-axis, and a Z-axis that are orthogonal to each other are illustrated in FIG. **2**. The accelerometer **200** is a sensor that detects acceleration in a direction along the Z-axis (hereinafter also referred to as a Z-axis direction) and includes a package **300** and an accelerometer element **400** accommodated in the package **300** as illustrated in FIG. **2**.
(13) The package **300** includes a base **310** having a recess **311** with an opening on a top surface and a plate-shaped lid **320** that is joined to the top surface of the base **310** and closes the opening of the recess **311**. An airtight space S is formed inside the package **300**, and the accelerometer element **400** is accommodated in the airtight space S.
(14) The accelerometer element **400** includes a substrate structure **500** fixed to the base **310**, the crystal oscillator **110** disposed on the substrate structure **500**, and a weight section **600** disposed on the substrate structure **500**.
(15) The substrate structure **500** is formed by a crystal substrate and forms a planar shape in an X-Y plane which is orthogonal to the Z-axis. Also, the substrate structure **500** includes a support section **510**, a movable section **520** that is to be displaced in the Z-axis direction with respect to the support section **510**, and a joint section **530** located between the support section **510** and the movable section **520** and is fixed to the base **310** at three places of the support section **510**. The movable section **520** is supported on one end by the support section **510** like a cantilever. A joint section **530**, which is thin with respect to the support section **510** and the movable section **520**, is disposed

between the support section **510** and the movable section **520**. The movable section **520** is likely to be displaced in the Z-axis direction with respect to the support section **510** by the joint section **530** functioning as a hinge.

(16) The crystal oscillator **110** is a twin-tuning-fork type oscillator formed by a crystal substrate. The crystal oscillator **110** also includes two oscillation beams **111** and **112**, a first base section **113** that terminates at one end of the two oscillation beams **111** and **112**, and a second base section **114** that terminates at the other end thereof. The oscillation beams **111** and **112** of the crystal oscillator **110** are disposed along the Y-axis. The crystal oscillator **110** is joined to the movable section **520** at the first base section **113** and joined to the support section **510** at the second base section **114**. The crystal oscillator **110** also includes a pair of excitation electrodes, which are not illustrated, disposed on the respective surfaces of the oscillation beams **111** and **112**. When an AC voltage drive signal is applied between the excitation electrodes, the oscillation beams **111** and **112** bend and oscillate in the X-axis direction so as to repeatedly move closer to and apart from each other in the X-axis direction.

(17) The weight section **600** is joined to the tip section of the movable section **520**. By disposing the weight section **600** on the movable section **520**, the mass of the movable section **520** is increased. Accordingly, the movable section **520** becomes easier to be displaced with small acceleration, and thus the resolution of the accelerometer **200** is improved.

(18) With the accelerometer **200**, it is possible to detect acceleration in the Z-axis direction as follows. When acceleration is applied to the accelerometer **200** in the Z-axis direction, the movable section **520** is displaced in the Z-axis direction with respect to the support section **510** with the joint section **530** as a fulcrum. Tensile stress or compressive stress is applied to the crystal oscillator **110** due to the displacement, and the resonant frequency of the crystal oscillator **110** changes in accordance with the applied stress. Specifically, when receiving acceleration in the positive Z-axis direction, the movable section **520** is displaced in the negative Z-axis direction with respect to the support section **510**. Thereby tensile stress is applied to the crystal oscillator **110** to increase the resonant frequency of the crystal oscillator **110**. On the other hand, when receiving acceleration in the negative Z-axis direction, the movable section **520** is displaced in the positive Z-axis direction with respect to the support section **510**. Thereby compressive stress is applied to the crystal oscillator **110** to decrease the resonant frequency of the crystal oscillator **110**. Accordingly, the accelerometer **200** is able to detect acceleration in the Z-axis direction in accordance with a change in the resonant frequency of the crystal oscillator **110**.

(19) In the above, a brief description has been given of the accelerometer **200**. However, the configuration of the accelerometer **200** is not particularly limited. Also, the configuration of the crystal oscillator **110** is not particularly limited.

(20) Returning back the description of the oscillator circuit **100**. The inverter **120** is coupled between the power source VDD and ground. The inverter **120** according to the present embodiment is a CMOS inverter. By using a CMOS inverter for the inverter **120**, it is possible to easily increase its gain. Also, the oscillator circuit **100** has low power consumption. However, as long as the inverter **120** performs its functions, the inverter **120** is not particularly limited.

(21) The feedback resistor **130** is coupled in parallel to the inverter **120**. In the oscillator circuit **100**, the feedback resistor **130** feeds back a signal from the output portion of the inverter **120** to continue the oscillation of the crystal oscillator **110**. Also, the limiting resistor **140** is coupled between the output end of the inverter **120** and the input end of the crystal oscillator **110**. In the oscillator circuit **100**, by using the limiting resistor **140**, it is possible to control the current flowing to the crystal oscillator **110**, adjust the negative resistor and the excitation level, and suppress the abnormal oscillation and the frequency fluctuations of the crystal oscillator **110**. Also, the first load capacitor **150** is coupled between the input end of the crystal oscillator **110** and ground, and the second load capacitor **160** is coupled between the output end of the crystal oscillator **110** and ground. In the oscillator circuit **100**, by using the first load capacitor **150** and the second load

capacitor **160**, it is possible to adjust the negative resistor, the excitation level, the oscillation frequency, and the like.

(22) Next, a detailed description will be given of the feedback resistor **130**. As illustrated in FIG. **1**, the feedback resistor **130** is constituted by a twin-T notch filter **131**, which is a band stop filter. Thereby, it becomes easy to configure the band stop filter. The twin-T notch filter **131** is constituted by three resistors R**1**, R**2**, and R**3**, and three capacitors C**1**, C**2**, and C**3**. Specifically, two resistors R**1** and R**2** coupled in series, and two capacitors C**2** and C**3** coupled in series are coupled in parallel. The capacitor C**1** is coupled between the resistors R**1** and R**2**, and ground. The resistor R**3** is coupled between the capacitors C**2** and C**3**, and ground.

(23) With the twin-T notch filter **131**, it is possible to obtain a desired center attenuation frequency Freq by adjusting the resistor values of the individual resistors R**1** to R**3** and the capacitance values of the individual capacitors C**1** to C**3**. For example, assuming that R**1**=R**2**=R, R**3**=R/2, C**2**=C**3**=C, and C**1**=C**2**, the attenuation becomes the maximum value when Freq=1/(2πRC). That is, with the twin-T notch filter **131**, it is possible to increase the impedance only in the vicinity of the center attenuation frequency Freq. On the other hand, the feedback resistor **130** needs to have an impedance equal to or higher than a threshold value It for the signal amplification using the inverter **120**.

(24) Therefore, as illustrated in FIG. **3**, the inverter **120** amplifies only the signal having frequency components included in a rejection band Sf which is a frequency band equal to or higher than the threshold value It. Accordingly, as illustrated in FIG. **3**, in the oscillator circuit **100**, the twin-T notch filter **131** is configured so that a resonant frequency fr of the crystal oscillator **110** is included in the rejection band Sf, more preferably, the resonant frequency fr is located at the center of the rejection band Sf. Thereby, the signal produced by amplifying only the signal having the frequency components of the resonant frequency fr and its vicinity are input from the inverter **120** to the crystal oscillator **110** via the limiting resistor **140**. The signal output from the crystal oscillator **110** is positively fed back to be input to the inverter **120**. In this regard, as described above, in the accelerometer **200**, the resonant frequency fr of the crystal oscillator **110** fluctuates due to the received acceleration, and thus in the present embodiment, the twin-T notch filter **131** is configured so that the full range of the fluctuations of the resonant frequency fr is included in the rejection band Sf.

(25) In this manner, by using the twin-T notch filter **131** for the feedback resistor **130**, it is possible to amplify only the signal of the frequency components included in the rejection band Sf. Therefore, it is possible to keep noise low compared with the case of configuring the feedback resistor **130** with a single resistor as the related art. Accordingly, the oscillator circuit **100** has excellent noise characteristics.

(26) In the above, a description has been given of the oscillator circuit **100**. As described above, the oscillator circuit **100** includes: the inverter **120**; the crystal oscillator **110** coupled in parallel to the inverter **120**; the first load capacitor **150** coupled to the input portion of the crystal oscillator **110**; the second load capacitor **160** coupled to the output portion of the crystal oscillator **110**; and the twin-T notch filter **131** coupled in parallel to the inverter **120** and working as a band stop filter having the rejection band Sf including the resonant frequency fr of the crystal oscillator **110**. With the oscillator circuit **100** having such a configuration, only the signal having frequency components included in the rejection band Sf is amplified by the inverter **120** and input to the crystal oscillator **110**. Accordingly, it is possible to keep noise low, and thus the oscillator circuit **100** has excellent noise characteristics.

(27) Also, as described above, the band stop filter is the twin-T notch filter **131**. Thereby, the configuration of the band stop filter becomes simple.

(28) Also, as described above, the inverter **120** is a CMOS inverter. Thereby, it is possible to easily increase its gain. Also, the oscillator circuit **100** has low power consumption.

(29) Also, as described above, the oscillator circuit **100** includes the limiting resistor **140** coupled

between the output portion of inverter **120** and the crystal oscillator **110**. Thereby, it is possible to control the current flowing into the crystal oscillator **110**, adjust the negative resistor and the excitation level, and suppress abnormal oscillation and frequency fluctuations of the crystal oscillator **110**.

Second Embodiment

(30) FIG. **4** is a circuit diagram of an oscillator circuit according to a second embodiment. FIG. **5** is a graph illustrating the characteristics of a twin-T notch filter. FIG. **6** and FIG. **7** are diagrams illustrating the corresponding waveforms of respective oscillation signals.

(31) An oscillator circuit **100** according to the present embodiment is the same as the above-described oscillator circuit **100** according to the first embodiment except that a first offset resistor **170** and a second offset resistor **180** are further included. In this regard, in the following description, regarding the oscillator circuit **100** according to the present embodiment, a description will be given mainly of the differences from those of the first embodiment described above, and a description will be omitted of the same matters. Also, in the drawings of the present embodiment, the same signs are added to the same components as those in the first embodiment.

(32) As illustrated in FIG. **4**, the oscillator circuit **100** according to the present embodiment further includes the first offset resistor **170** and the second offset resistor **180** that are coupled to the corresponding ends of the twin-T notch filter **131** in addition to the configuration of the first embodiment described above. The first offset resistor **170** is coupled between the output portion of the inverter **120** and the input portion of the twin-T notch filter **131**, and the second offset resistor **180** is coupled between the output portion of the twin-T notch filter **131** and the input portion of the inverter **120**. In the present embodiment, the first offset resistor **170** and the second offset resistor **180** have the same resistance value. However, the present disclosure is not limited to this, and they may have different resistance values with each other.

(33) The first and second offset resistors **170** and **180** have a function of offsetting the characteristics of the twin-T notch filter **131**. Specifically, as illustrate in FIG. **5**, the higher the resistance values of the first and second offset resistors **170** and **180**, the more the characteristics of the twin-T notch filter **131** are shifted upward without change. As the characteristics are shifted, the rejection band Sf is expanded as much as shifted. In this manner, by disposing the first and the second offset resistors **170** and **180** and adjusting the resistance values of the first and the second offset resistors **170** and **180**, it is possible to adjust the range of the rejection band Sf in any way, and thus it becomes easy to perform noise design.

(34) Also, the waveform of the oscillation signal output from the oscillator circuit **100** is determined by the overall value of phase noise and amplitude noise. The wider the rejection band Sf by increasing the resistance values of the first and second offset resistors **170** and **180**, the closer the oscillation signal comes to a rectangular wave as illustrated in FIG. **6**, and the lower the amplitude noise becomes. On the other hand, the narrower the rejection band Sf by decreasing the resistance values of the first and second offset resistors **170** and **180**, the closer the oscillation signal comes to a sine wave as illustrated in FIG. **7**, and the lower the phase noise becomes. Accordingly, by optimizing the resistance values of the first and second offset resistors **170** and **180**, it is possible to easily adjust the waveform in accordance with the characteristics of the frequency counter to be used, and to adjust the waveform so as to minimize the overall value of phase noise and amplitude noise.

(35) Further, it is possible to suppress overcurrent by disposing the first and second offset resistors **170** and **180**. Specifically, as illustrated in FIG. **5**, in the twin-T notch filter **131**, the further away from the center attenuation frequency Freq, the lower the impedance becomes. Accordingly, when a signal having a frequency component significantly away from the center attenuation frequency Freq is output from the inverter **120**, the output portion of the inverter **120** and the input portion thereof are short-circuited. In this regard, the “short circuit” includes the state of being coupled at low impedance. Accordingly, the output signal of the inverter **120** might be applied to the input

portion of the inverter **120** to cause an overcurrent, and thus the circuit might be damaged. To deal with this problem, by disposing the first and second offset resistors **170** and **180**, the current value input to the inverter **120** is limited, and it is possible to effectively suppress the damage of the circuit as described above.
(36) As described above, the oscillator circuit **100** according to the present embodiment includes: the first offset resistor **170** coupled between the output portion of the inverter **120** and the twin-T notch filter **131**; and the second offset resistor **180** coupled between the twin-T notch filter **131** and the input portion of the inverter **120**. In this manner, by disposing the first and second offset resistors **170** and **180**, it is possible to adjust the range of the rejection band Sf in any way, and thus noise design becomes easy. Also, by optimizing the first and second offset resistors **170** and **180**, it is possible to easily adjust the waveform in accordance with the characteristics of the frequency counter to be used, and to adjust the waveform so as to minimize the overall value of phase noise and amplitude noise. Also, it is possible to limit the value of the current input to the inverter **120**, and to effectively suppress the damage of the circuit.
(37) With the second embodiment as described above, it is possible to obtain the same advantages as those of the first embodiment described above.
(38) In the above, the descriptions have been given of the oscillator circuit according to the present disclosure in accordance with the embodiments with reference to the drawings. However, the present disclosure is not limited to these. It is possible to replace the configuration of each section with any configuration having the same function. Also, any other component may be added to the present disclosure. Also, the present disclosure may be made with a combination of any two components out of the individual embodiments described above.
(39) Also, in the embodiments described above, the twin-T notch filter **131** is used as a band stop filter. However, the present disclosure is not limited to this. For example, an active notch filter may be used, and a filter other than the notch filter may be used.
(40) Also, the embodiments described above include one inverter **120** and one twin-T notch filter **131** coupled in parallel to the inverter **120**. However, the present disclosure is not limited to this. For example, the present disclosure may have a configuration including a plurality of inverters **120** coupled in series and a plurality of twin-T notch filters **131** coupled in parallel to the corresponding inverters **120**.
(41) Also, in the embodiments described above, the crystal oscillator **110** is used as an accelerometer element for detecting acceleration. However, the present disclosure is not limited to this. For example, the crystal oscillator **110** may be used as an angular velocity sensor element for detecting angular velocity or may be used as an oscillation element that produces an oscillation signal oscillating at a predetermined frequency. Also, the crystal oscillator **110** may be used for an application other than these.

Claims

1. An oscillator circuit comprising: an inverter; a crystal oscillator coupled in parallel to the inverter; a first load capacitor coupled to an input portion of the crystal oscillator; a second load capacitor coupled to an output portion of the crystal oscillator; and a band stop filter coupled in parallel to the inverter and having a rejection band including a resonant frequency of the crystal oscillator.
2. The oscillator circuit according to claim 1, wherein the band stop filter is a twin-T notch filter.
3. The oscillator circuit according to claim 1, further comprising: a first offset resistor coupled between an output portion of the inverter and the band stop filter; and a second offset resistor coupled between the band stop filter and an input portion of the inverter.
4. The oscillator circuit according to claim 1, wherein the inverter is a CMOS inverter.

5. The oscillator circuit according to claim 1, further comprising a limiting resistor coupled between an output portion of the inverter and the crystal oscillator.